

ALI NIKKHAH

Open to Relocation & Remote | alinkkh9@gmail.com | +98 991 296 3951 | alinikkhah2001.github.io |
linkedin.com/in/alinikkhah2001

SUMMARY

Backend and systems engineer with production experience building high-concurrency async APIs, distributed streaming infrastructure, and edge-optimized ML serving pipelines. At Digikala, engineered the async FastAPI serving layer handling 5M+ daily queries, with multi-tier Redis caching, vLLM PagedAttention, and Triton Inference Server for LLM deployment. At Advanced Analytics Australia, delivered distributed video analytics infrastructure on AXIS edge hardware using Kafka, TensorRT, and Kubernetes. Full ownership of the DevOps lifecycle: Docker, Kubernetes, Terraform, ArgoCD, and full observability stacks in production.

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead (Backend & Systems)

Jan 2025 – Aug 2025

- Engineered the async **FastAPI** serving infrastructure handling over **5 million daily queries** (p99 <200ms); implemented multi-tier **Redis** caching and asynchronous request batching to fully isolate model execution latency from client-facing response times.
- Integrated **vLLM** with PagedAttention and **Triton Inference Server** for LLM and embedding model deployment, maximizing KV-cache utilization and GPU throughput under variable query burst patterns (**3.2x throughput gain, 60% cost reduction per 1K queries**).
- Designed a Modular Control Pipeline (MCP) decoupling text ingestion, chunking, retrieval, and synthesis phases, enabling zero-downtime model hot-swapping and live index migrations.
- Built stateful multi-agent routing networks using LangGraph cyclic directed graphs to manage multi-turn session state, dynamic context evaluation, and automated query rewriting loops.
- Supervised Kubernetes deployment of multi-tenant model serving clusters, isolating vision and language model compute nodes for independent horizontal scaling and fault containment (**200 pods, 40% GPU cost reduction**).

Advanced Analytics Australia

Remote Contract

Backend & Edge Systems Engineer

Sep 2025 – Apr 2026

- Built high-performance async Python streaming workers interfacing directly with AXIS network camera hardware, processing continuous live video feeds and routing them into a distributed **Apache Kafka** ingestion broker for fault-tolerant delivery under network spikes (**100+ cameras, 30 FPS, exactly-once semantics**).
- Compiled and quantized vision networks into **TensorRT** and **TF Lite** runtimes served via Triton Inference Server, eliminating Python-level inference overhead on resource-constrained edge nodes (**<50ms inference, 3.5x speedup vs Python**).
- Designed sensor calibration routines and signal-to-noise monitoring to maintain stable detection thresholds across indoor and outdoor deployment sites (**-35% false positives**).

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer

Sep 2025 – Mar 2026

- Developed async backend infrastructure in Python/FastAPI supporting high-frequency portfolio projection requests and automated asset trading execution; decoupled ingestion from core relational databases via multi-tier Redis caching and **Apache Kafka** message queuing (**<100ms p99, 10K req/s**).
- Maintained full system reliability and observability across microservices using **Prometheus**, **Grafana**, and the ELK log aggregation stack (**99.9% uptime, <1min MTTR**).

- Orchestrated complex **Airflow** DAGs and deployed **PeerDB** CDC replication clusters to continuously mirror transactional databases into OLAP engines with sub-minute latency (**¡60s CDC latency, 5TB/day**).

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Automated end-to-end model deployment pipelines with Docker, **ArgoCD**, and GitHub Actions, reducing release cycles to under 10 minutes; managed Kubernetes cluster orchestration and IaC via Terraform (**15+ models, 99.9% deploy success rate**).
- Built low-latency model inference APIs with FastAPI and TensorFlow Serving to handle concurrent traffic spikes; orchestrated distributed workloads with Ray and MLflow.

SELECTED PROJECTS & RESEARCH

Distributed Intelligent Surveillance Infrastructure

Advanced Analytics Australia

Core Backend & Edge Systems Architect

2025 – 2026

- Designed the full backend architecture for a multi-site industrial safety compliance system: async Python streaming workers feed a Kafka broker, which distributes camera frames to TensorRT/TF Lite inference containers served via Triton, with detection events written to a PostgreSQL audit store.
- Implemented dynamic sensor calibration and continuous SNR monitoring to maintain stable detection thresholds across variable lighting and exposure conditions at outdoor and indoor sites.

Enterprise Agentic AI Serving Infrastructure

Digikala

Lead Backend Engineer

2024 – 2025

- Architected the complete serving layer for a multi-agent RAG system at 5M+ daily queries: FastAPI front-end, vLLM/Triton model runtime with PagedAttention, multi-tier Redis cache, and LangGraph session-state management — all deployed on a Kubernetes multi-tenant cluster.

RESEARCH HIGHLIGHTS

University of British Columbia

Remote Research Collaboration

Research Assistant – Vision-Language Medical Imaging

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Remote Research Collaboration

Research Assistant – Video Motion Classification

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024